Two-way categorical tables

Lecture and worked examples

Two-way tables and marginal frequencies

A qualitative variable describes a category. It is nominal if its categories have no natural order (means of transport), ordinal if they are ordered (level of satisfaction). A two-way table counts individuals according to two variables. The sums of the rows and columns are the margins. A marginal frequency divides a margin by the total count.

Two-way tables and marginal frequencies

In a population of 100 individuals, two characteristics A, B are observed:

Worked answer: Two-way tables and marginal frequencies

$f(A) = 32/100 = 0.32$; $f(B) = 35/100 = 0.35$; $f(\overline{B}) = 65/100 = 0.65$. The frequency of $A \cap B$ is $20/100 = 0.20$; it is not a marginal frequency.

Two-way tables and marginal frequencies

A première class of 32 students is split as follows. Complete the margins and calculate the marginal frequencies “CS” and “girls”.

Worked answer: Two-way tables and marginal frequencies

$f(\mathrm{CS}) = 21/32 = 65.625\%$; $f(\mathrm{girls}) = 14/32 = 43.75\%$.

Conditional frequencies and reading sentences

The frequency of A among the individuals of B, if this subpopulation is nonempty, is

$$f_B(A) = \frac{\text{count satisfying } A \text{ and } B}{\text{count of } B}.$$

The denominator corresponds to the population introduced by “among”.

Conditional frequencies and reading sentences

In the CS table, calculate the frequency of girls among the students without CS, then the frequency of students without CS among the girls. Are the two results equal?

Worked answer: Conditional frequencies and reading sentences

Among the 11 students without CS, 5 are girls: $5/11 \approx 45.45\%$. Among the 14 girls, 5 do not take CS: $5/14 \approx 35.71\%$. The numerator is the same, the reference populations differ.

Conditional frequencies and reading sentences

In the table of 100 individuals, calculate $f_B(A)$, $f_A(B)$ and $f_{\overline{A}}(B)$. Translate each result into a sentence.

Worked answer: Conditional frequencies and reading sentences

$f_B(A) = 20/35 = 4/7 \approx 57.14\%$ of the individuals of B satisfy A. $f_A(B) = 20/32 = 62.5\%$ of the individuals of A satisfy B. $f_{\bar{A}}(B) = 15/68 \approx 22.06\%$ of the individuals not satisfying A satisfy B.

Conditional frequencies and reading sentences

Does “60% of cyclists are students” allow us to conclude that 60% of students are cyclists?

Worked answer: Conditional frequencies and reading sentences

No. For example, out of 10 cyclists, 6 are students; if the population has 100 students in all, only 6% of the students are cyclists.

Completing a table from frequencies

A group has 200 people, of whom 120 use a bus (B). Among these 120 people, 75% have a pass (A). Among the other 80, 25% have a pass.

1. Construct the two-way table $B/\overline{B}$ and $A/\overline{A}$ with its margins.
2. Calculate the marginal frequency of A.
3. Calculate the frequency of B among A.

Worked answer: Completing a table from frequencies

In B, $120 \times 0.75 = 90$ people satisfy A, so 30 do not. In $\overline{B}$, $80 \times 0.25 = 20$ satisfy A, and 60 do not.

	A	$\overline{A}$	Total
B	90	30	120
$\overline{B}$	20	60	80
Total	110	90	200

$f(A) = 110/200 = 55\%$. $f_A(B) = 90/110 = 9/11 \approx 81.82\%$.

Completing a table from frequencies

A headline announces “Three pass holders in four take the bus”, based on $f_B(A) = 75\%$. Explain the error and propose a correct sentence.

Worked answer: Completing a table from frequencies

The headline reverses the conditioning. The correct sentence is: “Three people in four among those who take the bus have a pass.” The proportion of pass holders who take the bus is here about 81.82%.

Python

```
# Exemple fictif : (transport, satisfaction)
donnees = [("bus", "oui"), ("marche", "oui"),
           ("bus", "non"), ("velo", "oui"),
           ("bus", "oui"), ("marche", "non")]

bus = [r for r in donnees if r[0] == "bus"]
bus_satisfaits = [r for r in donnees
                  if r[0] == "bus" and r[1] == "oui"]
autres = [r for r in donnees if not r[0] == "bus"]
```

Processing a file and collecting real data

Give the sizes of the three lists. Calculate the frequency of satisfaction among bus users. Replace `and` by `or` in the second filter: how many rows are then selected?

Worked answer: Processing a file and collecting real data

Sizes: $3, 2, 3$. Among bus users, the frequency of satisfaction is $2/3$. With `or`, we select the 3 bus users and the 2 satisfied people using another means: 5 rows, without counting twice the rows satisfying both criteria.

Processing a file and collecting real data

Collect anonymously two variables: main means of transport and journey time in ordered categories (< 15 min; from 15 to under 30 min; ≥ 30 min). One row per response, without names. Construct the two-way table, then two conditional frequencies in opposite directions. State the count, the collection date and any non-responses.

Processing a file and collecting real data

Carry out the collection and write two justified conclusions. Explain why the results do not automatically describe all students in Madagascar.

Worked answer: Processing a file and collecting real data

The values depend on the actual collection. Check: one category per variable and per response; sum of the cells equal to the number of usable responses; correct denominator for each frequency; announced rounding. The sample is one particular class, not a representative sample of all students in the country. Non-responses must be distinguished from an observed category.